RESEARCH PROPOSAL

Secure and Accurate Retrieval-Augmented Generation Framework for Domain-Adaptive Applications

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Aligned with ISRO's Space Vision 2047
Advancing autonomous AI systems for secure, evidence-based decision support across technical, operational, and administrative workflows.

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1. EXECUTIVE SUMMARY

This proposal presents a comprehensive plan to develop a **Secure and Accurate Retrieval-Augmented Generation (RAG) Framework** specifically designed for ISRO's operational needs. The system will provide intelligent, evidence-based assistance across technical, operational, and administrative domains while ensuring complete data sovereignty and security through offline, air-gapped deployment.

Key Innovation:

Unlike traditional RAG systems that rely on external APIs and cloud services, this framework is engineered for complete offline operation using state-of-the-art open-source models. It combines hybrid retrieval (dense and sparse embeddings), intelligent reranking, and rigorous grounding mechanisms to deliver accurate, verifiable responses with full citation traceability.

Strategic Value:

- **Enhanced Decision Support:** Enables rapid, accurate analysis of QA reports, procurement manuals, GFR rules, telemetry data, and technical documentation
- **Security by Design:** Air-gapped deployment ensures no data leaves ISRO infrastructure, with multi-layer security including MFA, RBAC, encryption, and comprehensive audit logging
- **Domain Adaptability:** Tailored for ISRO's diverse operational areas including technical specifications, administrative procedures, and quality assurance workflows
- **Knowledge Synthesis:** Automatically synthesizes information across multiple documents, identifying patterns, trends, and insights that would be time-consuming to discover manually
- **Operational Efficiency:** Reduces time spent searching through documentation from hours to seconds, allowing experts to focus on analysis and decision-making

Expected Outcomes:

Metric	Target	Impact
Retrieval Accuracy (Recall@10)	≥90%	High-quality document retrieval
Response Faithfulness	≥95%	Evidence-backed, hallucination-free outputs
Query Latency (P95)	≤3 seconds	Real-time user experience
Citation Accuracy	≥98%	Verifiable, traceable responses
User Satisfaction	≥85%	Practical, useful assistance

This proposal outlines a rigorous 10-week implementation plan with clear milestones, deliverables, and success criteria. The system will be production-ready, fully documented, and designed for long-term maintenance and evolution.

2. INTRODUCTION

2.1 Background and Context

The Indian Space Research Organisation (ISRO) generates and maintains vast repositories of technical documentation, operational procedures, quality assurance reports, administrative guidelines, and mission data. As ISRO advances toward its Space Vision 2047 objectives, the volume and complexity of this knowledge base continue to grow exponentially.

Current Challenges:

- **Information Fragmentation:** Critical knowledge is distributed across thousands of documents in various formats (PDFs, Word documents, spreadsheets, technical specifications)
- **Time-Consuming Searches:** Engineers and scientists spend significant time manually searching through documentation to find specific information, reducing time available for analysis and innovation
- **Knowledge Accessibility:** New team members face steep learning curves accessing institutional knowledge, and even experienced personnel may be unaware of relevant information in other domains
- **Quality Assurance Analysis:** Identifying patterns and trends in QA reports requires manual review of hundreds of documents, making it difficult to proactively address systemic issues
- **Policy Compliance:** Ensuring compliance with GFR rules and procurement procedures requires constant reference to complex documentation, increasing risk of errors
- **Security Requirements:** Sensitive ISRO data cannot be processed using external cloud-based AI services, limiting access to modern AI capabilities

The AI Revolution in Knowledge Management:

Recent advances in Large Language Models (LLMs) and Retrieval-Augmented Generation have demonstrated remarkable capabilities in understanding and synthesizing information from large document collections. However, most commercial solutions require cloud connectivity and raise significant security concerns for sensitive organizational data.

This proposal addresses these challenges by designing a RAG system that combines cutting-edge AI capabilities with stringent security requirements, enabling ISRO to leverage modern AI while maintaining complete data sovereignty.

2.2 Problem Statement

ISRO requires an intelligent knowledge management system that can:

- Accurately retrieve relevant information from diverse document types across multiple domains
- Generate precise, evidence-backed responses with full citation traceability
- Operate completely offline in air-gapped environments for maximum security
- Adapt to different operational contexts (technical, administrative, analytical)

- Prevent hallucinations and ensure factual accuracy in all outputs
- Support multiple Indian languages for inclusive accessibility
- Scale to handle millions of documents without performance degradation
- Provide comprehensive audit trails for compliance and security

Technical Complexity:

Building such a system presents significant technical challenges. Standard RAG implementations often suffer from poor retrieval quality, hallucinated responses, lack of citation accuracy, and inability to handle domain-specific terminology. Furthermore, achieving production-grade performance while maintaining offline operation requires sophisticated optimization and careful technology selection.

2.3 Objectives

This project aims to develop a production-ready RAG framework with the following specific objectives:

Primary Objectives:

- **High-Accuracy Retrieval:** Achieve ≥90% Recall@10 through hybrid retrieval combining dense embeddings (BGE-M3) and sparse methods (BM25), with intelligent reranking
- **Factual Accuracy:** Maintain ≥95% faithfulness score ensuring all generated statements are grounded in retrieved evidence, with hallucination rate <2%
- **Domain Adaptation:** Support technical, administrative, and QA analysis domains with customized retrieval strategies and prompt templates for each
- **Security Implementation:** Deploy complete air-gapped system with multi-factor authentication, role-based access control, encryption at rest and in transit, and comprehensive audit logging
- **Performance Optimization:** Deliver sub-3-second P95 latency for end-to-end query processing while supporting 20+ queries per second throughput

Secondary Objectives:

- Develop comprehensive evaluation framework with automated quality monitoring
- Create detailed documentation and training materials for system administrators and end users
- Establish continuous improvement processes for ongoing optimization
- Design modular architecture enabling future enhancements and integrations

3. METHODOLOGY

3.1 RAG Framework Architecture

The proposed RAG framework implements a sophisticated multi-stage pipeline optimized for accuracy, security, and performance. The architecture is designed around six core layers:

Layer 1: Document Ingestion and Processing

The system processes diverse document formats through a unified pipeline:

- **Format Detection:** Automatically identifies document type (PDF, DOCX, XLSX, HTML, Markdown, images)
- **Content Extraction:** Uses specialized parsers (PyMuPDF for PDFs, python-docx for Word documents, pandas for spreadsheets, DocTR/Tesseract for OCR)
- **Metadata Enrichment:** Extracts and stores document metadata (domain classification, date, author, security classification, source system)
- **Quality Validation:** Checks text extraction quality, flags low-confidence OCR results, validates document structure
- **Chunking Strategy:** Implements domain-adaptive chunking (512 tokens for general text, larger chunks for tables/code, semantic chunking for coherent sections)

Layer 2: Hybrid Indexing System

To maximize retrieval quality, the system employs dual indexing:

Index Type	Purpose	Technology	Strength
Dense Vector	Semantic similarity	FAISS + BGE-M3 embeddings	Captures meaning, synonyms, context
Sparse Lexical	Keyword matching	BM25 algorithm	Exact term matches, technical jargon

Results from both indices are combined using Reciprocal Rank Fusion, which has been shown to outperform either method alone by 15-20% in retrieval accuracy.

Layer 3: Multi-Stage Retrieval Pipeline

- **Stage 1 - Query Understanding:** Analyzes query intent, classifies domain (technical/administrative/QA), performs query expansion for acronyms and synonyms
- **Stage 2 - Candidate Retrieval:** Executes hybrid search retrieving top-100 documents from both dense and sparse indices
- **Stage 3 - Reranking:** Applies BGE-Reranker-v2-M3 cross-encoder to precisely score query-document relevance, reducing to top-10 highest-quality results
- **Stage 4 - Context Grounding:** Validates retrieved documents for relevance, extracts citation information, checks for contradictions, assembles final context window

Layer 4: Generation and Synthesis

The generation layer uses Meta Llama 3.1-8B-Instruct (128K context window) with carefully engineered prompts that enforce:

- Answer only from provided context (strict grounding)
- Include citations for every factual claim
- Acknowledge uncertainty when evidence is ambiguous or insufficient
- Maintain domain-appropriate technical language and precision
- Structure responses for clarity (executive summary, detailed explanation, citations)

Layer 5: Validation and Quality Control

Before presenting responses to users, the system performs multi-level validation:

- **Faithfulness Verification:** Uses NLI (Natural Language Inference) models to verify every claim is entailed by retrieved context
- **Citation Validation:** Confirms all citations accurately reference source documents and specific passages
- **Hallucination Detection:** Employs SelfCheckGPT and consistency checking to identify potentially fabricated information
- **Security Screening:** Checks for PII leakage, classification violations, prompt injection attempts
- **Quality Scoring:** Assigns confidence scores based on evidence strength, consistency, and completeness

Layer 6: Monitoring and Continuous Improvement

The system continuously monitors performance across multiple dimensions:

- Retrieval metrics (Recall@K, NDCG, MRR, Precision)
- Generation quality (Faithfulness, Answer Relevance, Context Precision)
- System performance (Latency, Throughput, Resource Utilization)
- Security events (Authentication failures, Access violations, Anomalies)
- User feedback (Satisfaction scores, Query reformulation rates, Manual overrides)

Automated alerts trigger when metrics deviate from targets, enabling rapid response and continuous optimization.

3.2 Technical Approach

Embedding Model Selection: BGE-M3

The system uses BAAI BGE-M3 (BAAI General Embedding - Multilingual) as the primary embedding model. This choice is driven by several critical factors:

- **State-of-the-art Performance:** MTEB score of 66.1, outperforming most alternatives on retrieval tasks
- **Multilingual Support:** Trained on 100+ languages including Hindi, Tamil, Telugu - critical for ISRO's diverse documentation and future inclusivity goals
- **Long Context:** 8192 token maximum length handles complex technical documents without truncation
- **Multi-functionality:** Produces dense, sparse, and multi-vector representations in a single model
- **Offline Operation:** MIT license, fully downloadable, no API dependencies
- **Efficient:** 568M parameters - runs efficiently on standard GPU hardware (2GB VRAM)

Large Language Model: Llama 3.1-8B-Instruct

Meta's Llama 3.1-8B-Instruct serves as the generation backbone:

- **Instruction Following:** Specifically fine-tuned to follow complex instructions, essential for structured RAG prompts requiring citations and grounding
- **Extended Context:** 128,000 token context window accommodates large retrieved document sets
- **Reasoning Capability:** MMLU score of 68.4 demonstrates strong analytical and reasoning abilities
- **Quantization Support:** 4-bit quantization enables deployment on 16GB GPU while maintaining quality
- **Tool Use:** Native support for structured outputs and function calling for future extensibility

Alternative LLM Strategy:

For maximum accuracy on technical/mathematical content, the system supports optional deployment of Qwen 2.5-7B-Instruct (MMLU 70.3, stronger on technical reasoning). Domain routing can automatically select the optimal model based on query classification.

Reranking: BGE-Reranker-v2-M3

Cross-encoder reranking dramatically improves retrieval precision:

- Evaluates query-document pairs jointly rather than independently
- Typically improves NDCG by 15-20% over embedding-only retrieval
- Fast inference (100ms for 20 documents) enables real-time use
- Multilingual support matching embedding model

Vector Database: FAISS

Facebook AI Similarity Search (FAISS) provides the vector storage and search engine:

- **Offline-First:** Pure library (not a service), operates entirely locally with no network dependencies
- **Performance:** Sub-10ms search times on millions of vectors through optimized index structures
- **Scalability:** Proven at billion-scale deployments (used by Meta, OpenAI)
- **Flexibility:** Multiple index types (Flat, IVF, HNSW, PQ) for different scale/accuracy tradeoffs
- **GPU Acceleration:** Optional CUDA support for even faster search
- **Backup Simplicity:** Indices are simple files enabling trivial backup/restore

Inference Optimization: vLLM

vLLM (Very Large Language Model) engine provides production-grade LLM serving:

- **PagedAttention:** Novel algorithm achieving up to 24x higher throughput than naive implementation
- **Dynamic Batching:** Automatically batches concurrent requests for maximum GPU utilization
- **OpenAI-Compatible API:** Standard interface simplifying integration and future model swaps
- **Low Latency:** Optimized KV cache management minimizes response time

3.3 Domain Adaptation Strategy

ISRO's operational domains have distinct characteristics requiring tailored approaches:

Technical Domain

Characteristics: Precise specifications, mathematical formulas, technical jargon, tables of numerical data

- **Retrieval Optimization:** Higher weight on exact term matching (BM25) to capture specific part numbers, codes, and technical terminology
- **Chunking Strategy:** Preserve table structures, keep related specifications together, use semantic boundaries (sections/subsections)
- **Prompt Engineering:** Emphasize precision, include units in responses, request step-by-step calculations when appropriate
- **Validation:** Extra scrutiny on numerical values, unit conversions, formula correctness

Administrative Domain

Characteristics: Policy documents, procedures, compliance rules, hierarchical guidelines

- **Retrieval Optimization:** Balance semantic understanding with rule citation accuracy
- **Citation Requirements:** Mandatory rule/section number citations for all policy statements
- **Prompt Engineering:** Request structured responses (eligibility criteria, procedures, exceptions), highlight relevant rules
- **Validation:** Zero tolerance for hallucination - verify every rule citation exists, check for policy contradictions

QA Analysis Domain

Characteristics: Issue reports, root cause analysis, trend identification, corrective actions

- **Retrieval Optimization:** Support temporal queries (issues in date ranges), enable aggregation across multiple reports
- **Analysis Capabilities:** Identify recurring patterns, correlate issues across subsystems, track corrective action effectiveness
- **Prompt Engineering:** Request trend analysis, comparative summaries, risk assessments
- **Metadata Utilization:** Leverage timestamps, severity levels, affected systems for targeted retrieval

Domain Routing

The system automatically classifies queries by domain using a lightweight classifier trained on example queries. This enables automatic selection of domain-appropriate retrieval strategies, prompt templates, and validation criteria without requiring users to manually specify the domain.

4. APPROACH AND IMPLEMENTATION

4.1 Technology Stack

The system is built entirely on open-source technologies enabling offline deployment:

Component	Technology	Version	Purpose
Embedding	BGE-M3	Latest	Semantic search
LLM	Llama 3.1-8B-Instruct	Latest	Response generation
Reranker	BGE-Reranker-v2-M3	Latest	Result refinement
Vector DB	FAISS	1.8.0+	Vector storage/search
Lexical Search	BM25	rank-bm25	Keyword matching
Inference Engine	vLLM	Latest	LLM serving
API Framework	FastAPI	0.108+	REST API
Database	PostgreSQL 15	+ pgvector	Metadata storage
PDF Processing	PyMuPDF	1.23+	PDF parsing
OCR	DocTR + Tesseract	Latest	Scanned docs
Containerization	Docker	Latest	Deployment
Monitoring	Prometheus + Grafana	Latest	Observability
Language	Python	3.11+	Implementation

Deployment Architecture

The system deploys as a containerized application with the following structure:

- **API Layer:** FastAPI application handling user requests, authentication, and routing
- **Retrieval Service:** FAISS vector search and BM25 keyword matching
- **Reranking Service:** BGE-Reranker for precision improvement
- **Generation Service:** vLLM serving Llama 3.1 model
- **Database Layer:** PostgreSQL for metadata, user data, and audit logs
- **Monitoring Stack:** Prometheus metrics collection, Grafana dashboards

All components run in isolated Docker containers orchestrated via Docker Compose (or Kubernetes for larger deployments), enabling easy scaling, updates, and disaster recovery.

Hardware Requirements

Component	Minimum Specification	Recommended Specification
CPU	16 cores (Xeon/EPYC)	32 cores (Xeon Gold/EPYC)
RAM	64GB DDR4	128GB DDR4

GPU	1x NVIDIA A10 (24GB)	1x NVIDIA A100 (40GB)
Storage	1TB NVMe SSD	2TB NVMe SSD (RAID 1)
Network	10 Gbps internal	10 Gbps internal

Note: The recommended specification supports 100K-1M documents with <3 second response times and 20+ concurrent users. Hardware costs estimated at ■8-10 lakhs (one-time investment).

4.2 Security Architecture

Security is foundational to the system design, implementing defense-in-depth across six layers:

Layer 1: Network & Infrastructure Security

- **Air-Gapped Deployment:** Complete physical isolation from external networks, no internet connectivity
- **Network Segmentation:** Internal VLANs separating application, data, and management layers
- **Infrastructure Hardening:** Minimal OS installation, disabled unnecessary services, regular offline patching
- **Physical Security:** Server room access controls, biometric entry, 24/7 monitoring

Layer 2: Authentication & Access Control

- **Multi-Factor Authentication:** Smart card + PIN + biometric (fingerprint/iris) for all logins
- **Role-Based Access Control (RBAC):** Predefined roles (Admin, Expert, Analyst, Viewer, Auditor) with granular permissions
- **Attribute-Based Access Control (ABAC):** Dynamic policies based on time, location, document classification
- **Session Management:** JWT tokens with 30-minute expiration, automatic re-authentication for sensitive ops

Layer 3: Data Encryption

- **At Rest:** AES-256-GCM encryption for all data stores, HSM-based key management, quarterly key rotation
- **In Transit:** TLS 1.3 for all internal communications, mutual certificate authentication
- **Database-Level:** Transparent Data Encryption (TDE) for PostgreSQL, field-level encryption for PII
- **Vector Indices:** Encrypted FAISS indices with separate encryption keys per domain

Layer 4: Query & Response Security

- **Input Validation:** Query length limits, character whitelisting, injection pattern detection
- **Prompt Injection Detection:** Pattern matching and LLM-based analysis to identify adversarial inputs
- **Output Filtering:** PII detection and redaction, classification-level checking, response length limits
- **Rate Limiting:** Per-user query limits (100/hour), anomaly detection for unusual patterns

Layer 5: Audit & Compliance

- **Comprehensive Logging:** All authentication events, queries, document access, system changes
- **Immutable Logs:** Write-once-read-many storage, encrypted log archival, 7-year retention
- **Real-Time Monitoring:** SIEM integration, automated alerting, security dashboard
- **Compliance Reporting:** Monthly access reports, quarterly security assessments, audit trail reconstruction

Layer 6: Data Governance

- **Model Version Control:** Git-based versioning, checksum verification, rollback capability
- **Data Provenance:** Complete lineage tracking from source to retrieval
- **Backup & Recovery:** Daily encrypted backups, offsite storage, quarterly DR drills, 4-hour RTO
- **Configuration Management:** Infrastructure as Code, change approval workflows, audit trail

4.3 Evaluation Framework

The system implements comprehensive quality monitoring across six metric categories:

Category 1: Retrieval Performance

Metric	Formula	Target	Purpose
Recall@10	Relevant in top-10 / Total relevant	≥0.90	Find all important docs
NDCG@10	DCG / Ideal DCG	≥0.88	Ranking quality
MRR	1 / First relevant rank	≥0.85	Top result quality
Precision@5	Relevant in top-5 / 5	≥0.75	Result relevance

Category 2: Generation Quality

Metric	Measurement	Target	Purpose
Faithfulness	Claims supported / Total claims	≥0.95	Prevent hallucination
Answer Relevance	Semantic similarity to query	≥0.80	Address user question
Context Precision	Relevant context used / Retrieved	≥0.70	Efficient retrieval
Context Recall	Ground truth in context / Total	≥0.85	Complete information

Category 3: Hallucination Detection

- **Hallucination Rate:** ≤2% - Percentage of responses containing unsupported claims
- **Citation Accuracy:** ≥98% - Percentage of citations correctly referencing sources
- **Factual Consistency:** ≥0.90 NLI score - Logical entailment from context
- **Knowledge Boundary:** ≥95% - Proper refusal when lacking sufficient information

Category 4: System Performance

- **Query Latency (P95):** ≤3 seconds - 95th percentile end-to-end response time
- **Throughput:** ≥20 queries/second - Concurrent query handling capacity
- **Resource Utilization:** CPU <70%, GPU <80%, RAM <85% - Headroom for spikes
- **Availability:** ≥99.9% - System uptime (excluding planned maintenance)

Category 5: Domain-Specific Accuracy

- **Technical Domain:** ≥92% accuracy on specifications, calculations, technical queries
- **Administrative Domain:** ≥95% accuracy on policies, procedures, compliance rules
- **QA Analysis Domain:** ≥90% accuracy on issue identification, trend analysis
- **Cross-Domain Consistency:** ≥88% - Consistent answers across related queries

Category 6: User Experience

- **User Satisfaction:** ≥85% thumbs up rating on responses
- **Task Completion:** ≥80% - Users get satisfactory answers without escalation

- **Query Reformulation:** ≤20% - Users don't need to rephrase due to poor results
- **Citation Engagement:** ≥30% - Users click through to source documents

Automated Alerting

The system monitors all metrics in real-time and triggers alerts at three severity levels:

- **CRITICAL (5-min SLA):** Hallucination rate >5%, System down, Security breach, Faithfulness <0.85
- **HIGH (1-hour SLA):** P95 latency >5s, Resource >90%, User satisfaction <0.75, Citation accuracy <0.90
- **MEDIUM (4-hour SLA):** Recall <0.80, Error rate >2%, Unusual access patterns

5. EXPECTED DELIVERABLES

The project will deliver a complete, production-ready system with comprehensive documentation:

Primary Deliverables

End-to-End RAG Framework

Complete source code in Python with modular architecture, offline-compatible, optimized for ISRO's requirements

Deployed System

Fully operational instance running on ISRO infrastructure with all security controls active

Multi-Domain Implementations

Configured and tested for technical, administrative, and QA analysis domains

Pre-trained Models

Downloaded and verified BGE-M3, Llama 3.1, BGE-Reranker models ready for offline use

Evaluation Dataset

1100+ test queries across domains with ground truth annotations for ongoing validation

Documentation Deliverables

- **System Architecture Document:** Detailed technical design, component interactions, data flows
- **Installation Guide:** Step-by-step setup instructions with troubleshooting
- **Administrator Manual:** System configuration, user management, backup/restore procedures
- **User Guide:** How to query the system, interpret results, provide feedback
- **API Documentation:** Complete REST API reference with examples
- **Security Operations Manual:** Security policies, incident response, audit procedures
- **Evaluation Report:** Baseline metrics, benchmark results, comparison with targets

Training Materials

- Video tutorials for end users (query techniques, result interpretation)
- Administrator training sessions (system maintenance, troubleshooting)
- Security team briefings (threat models, monitoring, incident response)
- Developer documentation (code structure, extension points, best practices)

Monitoring & Analytics

- Grafana dashboards for real-time system monitoring
- Prometheus alerting rules configured for all critical metrics
- Custom evaluation scripts for ongoing quality assessment

- Usage analytics reports (query patterns, user behavior, system load)

6. IMPLEMENTATION ROADMAP

The project follows a structured 10-week implementation plan with clear milestones:

Phase	Duration	Key Activities	Deliverables
Phase 1: Foundation	Weeks 1-2	• Environment setup • Install dependencies • Download models • Basic retrieval system • Document parsing	• Working dev environment • Vector + BM25 retrieval • PDF/DOCX processing
Phase 2: RAG Pipeline	Weeks 3-4	• Hybrid retrieval • Reranking integration • vLLM setup • Prompt engineering • Citation system	• Complete RAG pipeline • Domain prompt templates • End-to-end testing
Phase 3: Domain Adaptation	Weeks 5-6	• Ingest ISRO docs • Domain-specific tuning • Technical domain testing • Admin domain testing • QA domain testing	• Domain configurations • Ingested documentation • Domain eval results
Phase 4: Security & Deploy	Weeks 7-8	• MFA implementation • RBAC setup • Encryption config • Audit logging • Docker deployment • Monitoring setup	• Secured system • Deployed instance • Monitoring dashboards • User documentation
Phase 5: Evaluation & Optimize	Weeks 9-10	• Create test dataset • Run evaluations • Identify issues • Optimize performance • Final testing • Documentation	• Evaluation report • Optimized system • Complete documentation • Training materials

Milestone Reviews

- **Week 2:** Demo basic retrieval system, review architecture decisions
- **Week 4:** Demo end-to-end RAG pipeline, evaluate response quality
- **Week 6:** Present domain-specific results, discuss optimization needs
- **Week 8:** Security audit, deployment verification, user acceptance testing
- **Week 10:** Final presentation with comprehensive evaluation results

7. RESOURCE REQUIREMENTS

7.1 Hardware Infrastructure

The recommended hardware configuration for 100K-1M documents:

Component	Specification	Quantity	Estimated Cost (■)
Server (CPU)	32-core Xeon Gold/EPYC, 128GB RAM	1	3,00,000
GPU	NVIDIA A100 40GB	1	4,50,000
Storage	2TB NVMe SSD (Enterprise, RAID 1)	1 set	80,000
Networking	10Gbps switch + cables	1	50,000
UPS	3KVA online UPS	1	60,000
Rack & Cooling	42U rack with cooling	1	40,000
	Total Hardware Cost		9,80,000

Note: All costs are approximate. Hardware is a one-time investment with no recurring licensing fees.

7.2 Software & Licenses

All software components are open-source with permissive licenses:

- BGE-M3, BGE-Reranker: MIT License (fully permissive)
- Llama 3.1: Llama 3 Community License (permissive for research/commercial use)
- FAISS, vLLM, FastAPI, PostgreSQL: MIT/Apache 2.0/PostgreSQL License
- Python ecosystem: Python Software Foundation License
- Operating System: Ubuntu 24.04 LTS (free, open-source)
- **Total Software Cost: ■0** (no licensing fees)

7.3 Human Resources

Role	Expertise Required	Time Commitment
ML Engineer	RAG systems, Python, LLMs	Full-time (10 weeks)
DevOps Engineer	Docker, system administration	Part-time (4 weeks)
Security Engineer	Infrastructure security, RBAC	Part-time (4 weeks)
Domain Experts	ISRO operations knowledge	Consultation (ongoing)
Project Manager	Coordination, reporting	Part-time (10 weeks)

7.4 Total Project Cost Summary

Hardware Infrastructure: ■9,80,000 (one-time)

Software Licenses: ■0 (open-source)

Personnel (if outsourced): ■15-20 lakhs (10 weeks)

Contingency (10%): ■2,50,000

Estimated Total: ■27-32 lakhs

Note: If implemented using internal ISRO resources, personnel costs may be significantly reduced.

8. RISK ASSESSMENT AND MITIGATION

Risk	Probability	Impact	Mitigation Strategy
Model hallucination in critical domains	Medium	High	• Multi-layer validation (NLI, SelfCheckGPT) • Strict grounding prompts • Human review for critical queries • Automated hallucination detection
Insufficient retrieval accuracy	Medium	High	• Hybrid retrieval (dense + sparse) • Reranking with cross-encoder • Continuous evaluation with test sets • Domain-specific optimization
Performance degradation with scale	Low	Medium	• Optimized FAISS indices (HNSW/IVF) • vLLM for efficient inference • Horizontal scaling capability • Performance monitoring & alerting
Security vulnerabilities	Low	High	• Defense-in-depth architecture • Regular security audits • Penetration testing • Prompt injection detection • Comprehensive access logging
Data quality issues in source documents	Medium	Medium	• OCR quality validation • Document preprocessing pipeline • Manual quality spot-checks • User feedback mechanism
User adoption challenges	Medium	Medium	• Comprehensive training program • Intuitive user interface • Gradual rollout with pilot users • Continuous UX improvements
Hardware failures	Low	High	• RAID storage configuration • Regular backups (daily) • Disaster recovery procedures • Spare hardware components

Risk Monitoring

All identified risks will be monitored throughout the project lifecycle with weekly risk review meetings and documented in a risk register. Mitigation strategies will be updated as new risks emerge or existing risks evolve.

9. CONCLUSION

This proposal presents a comprehensive plan to develop a secure, accurate, and production-ready Retrieval-Augmented Generation framework specifically designed for ISRO's unique operational requirements. By combining state-of-the-art open-source AI models with rigorous security controls and domain adaptation strategies, the system will provide intelligent, evidence-based assistance while maintaining complete data sovereignty.

Key Differentiators:

- **Security-First Design:** Air-gapped deployment with multi-layer security ensures no data leaves ISRO infrastructure, addressing critical security requirements that eliminate most commercial alternatives
- **Accuracy Through Validation:** Multi-stage validation pipeline including faithfulness verification, citation accuracy checks, and hallucination detection ensures reliable, verifiable outputs
- **Domain Intelligence:** Tailored retrieval strategies and prompt templates for technical, administrative, and QA analysis domains maximize relevance and usefulness
- **Production-Ready Engineering:** Built on battle-tested open-source technologies (FAISS, vLLM, PostgreSQL) with comprehensive monitoring, alerting, and disaster recovery
- **Zero Vendor Lock-In:** Complete control over all components, no proprietary dependencies, all code and models owned by ISRO

Strategic Impact:

The successful implementation of this RAG framework will position ISRO at the forefront of AI-enabled knowledge management in the aerospace sector. By automating routine information retrieval and synthesis tasks, the system will free expert personnel to focus on high-value analysis, innovation, and decision-making. The offline, secure architecture ensures ISRO can leverage cutting-edge AI capabilities while maintaining the stringent security posture required for sensitive space operations.

Long-Term Vision:

This project establishes foundational infrastructure that can evolve with ISRO's growing needs. The modular architecture supports future enhancements including:

- Integration with mission planning systems for real-time operational support
- Expansion to additional languages for increased accessibility
- Agent capabilities for complex multi-step analytical workflows
- Integration with telemetry databases for automated anomaly analysis
- Knowledge graph construction for advanced reasoning and discovery

The proposed RAG framework represents a significant advancement in how ISRO manages and leverages its vast knowledge assets. We are confident that this system will deliver measurable improvements in operational efficiency, decision quality, and knowledge accessibility while meeting the highest standards of security and reliability required for ISRO's mission-critical operations.

We respectfully submit this proposal for consideration and look forward to the opportunity to contribute to ISRO's Space Vision 2047 through advanced AI-enabled knowledge systems.

APPENDIX A: DETAILED TECHNICAL SPECIFICATIONS

A.1 Model Specifications

Model	Parameters	Context	License	Hardware Req
BGE-M3	568M	8192 tokens	MIT	2GB VRAM
Llama 3.1-8B	8B	128K tokens	Llama 3	16GB VRAM (FP16)
BGE-Reranker	568M	8192 tokens	MIT	2GB VRAM

A.2 Performance Benchmarks

Metric	Baseline Target	Stretch Goal
Retrieval Recall@10	0.90	0.95
Faithfulness Score	0.95	0.98
P95 Query Latency	3.0s	2.0s
Hallucination Rate	2%	<1%
User Satisfaction	85%	90%

APPENDIX B: REFERENCES

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